

Immunosuppression: An Overview of Infection Risk

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Introduction & Epidemiology

The growing population of immunocompromised hosts

- **Estimated 5-6% of the Italian population** is immunocompromised (Martinson and Lapham, 2024)
- **1%** of children fall into cohort of immunocompromised (VERDI project)
- **2.8%** meet criteria for drug-induced immunosuppression (Wallace et al., 2021)

What causes immunocompromise?

Major categories:

- Active treatment (chemotherapy) for malignancies
- Solid organ transplant (SOT)
- Hematopoietic cell transplant (HCT)
- CAR-T cell therapy
- Primary immunodeficiency
- Advanced HIV infection
- High-dose corticosteroids & biologics

The net state of immunosuppression

Defining net state of immunosuppression

A concept by Dr. Robert Rubin (Massachusetts General Hospital-Harvard, Boston):
“Father” of transplant infectious diseases”



“Composite of host factors, underlying disease, treatment, and other factors contributing to infection risk”

This remains one of the most useful conceptual frameworks for thinking about infection risk in immunocompromised patients. It

Components of the “net immunosuppressed state”

Host Factors

- Advanced age
- Malnutrition
- Diabetes
- Organ dysfunction
- Hypogammaglobulinemia

Treatment Factors

- Immunosuppressive drugs
- Chemotherapy
- Radiation
- Surgery/hardware
- Duration of therapy

Components (continued)

Underlying Disease

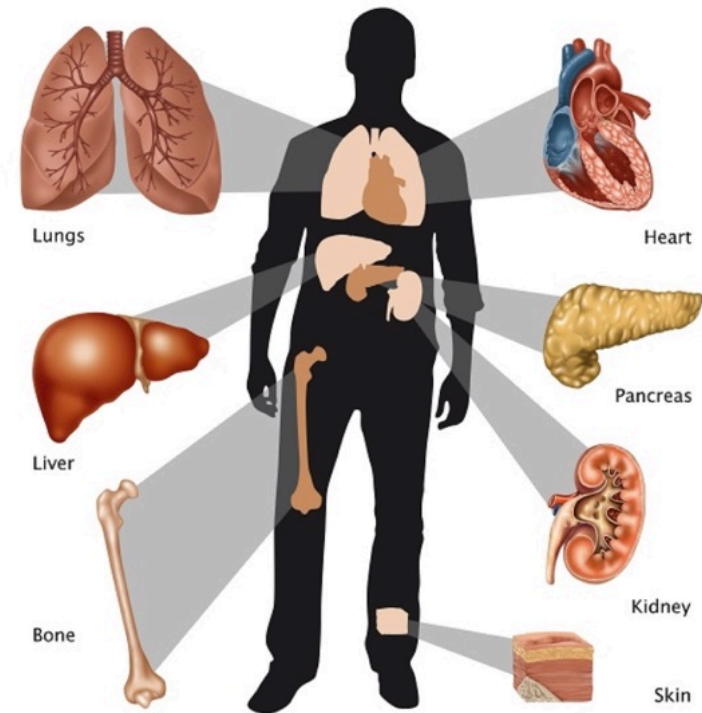
- Autoimmune disease
- Malignancy type
- Organ failure stage

Infectious Factors

- HIV, CMV, EBV status
- Microbiome alterations
- Prior infections

SOT Recipients

- **6%** died from infection within first year (Swiss cohort) (Delden et al., 2020)
- **55%** had infections in first year (German renal cohort) (Sommerer et al., 2022)
 - Half occurred in first 3 months
 - Bacteria: 66%, Viruses: 29%, Fungi: 5%

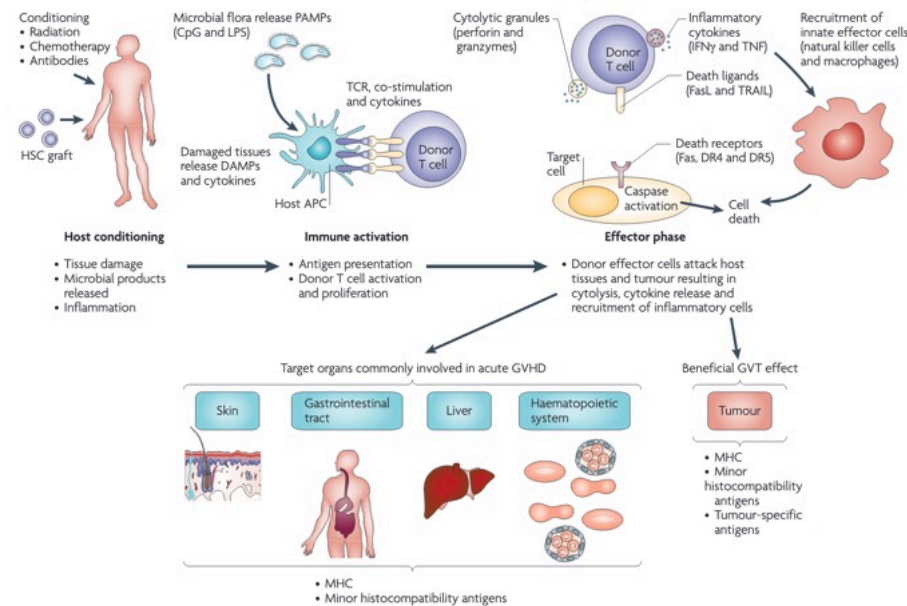


SOT recipients face particular risks from surgical complications, hardware-related infections, and the combination of surgical stress plus

Hematopoietic stem cell transplantation (HSCT) recipients

Type	Stem Cell Source	Donor	Immunosuppression
Autologous	Peripheral blood, Bone marrow	Self	Moderate; no GvHD prophylaxis required; recovery within weeks
Allogeneic — matched related	Peripheral blood, Bone marrow, Umbilical cord blood	HLA-matched sibling or family member	Severe; prolonged due to GvHD prophylaxis and risk of GvHD
Allogeneic — matched unrelated (MUD)	Peripheral blood, Bone marrow, Umbilical cord blood	HLA-matched unrelated donor (registry)	Very severe; higher GvHD risk than matched related; intensive prophylaxis
Allogeneic — haploidentical	Peripheral blood, Bone marrow	Half-matched family member (parent, child, sibling)	Very severe; requires intensive T-cell depletion or post-transplant cyclophosphamide
Allogeneic — umbilical cord blood	Umbilical cord blood	Unrelated cord blood unit	Very severe; delayed immune reconstitution due to low cell dose

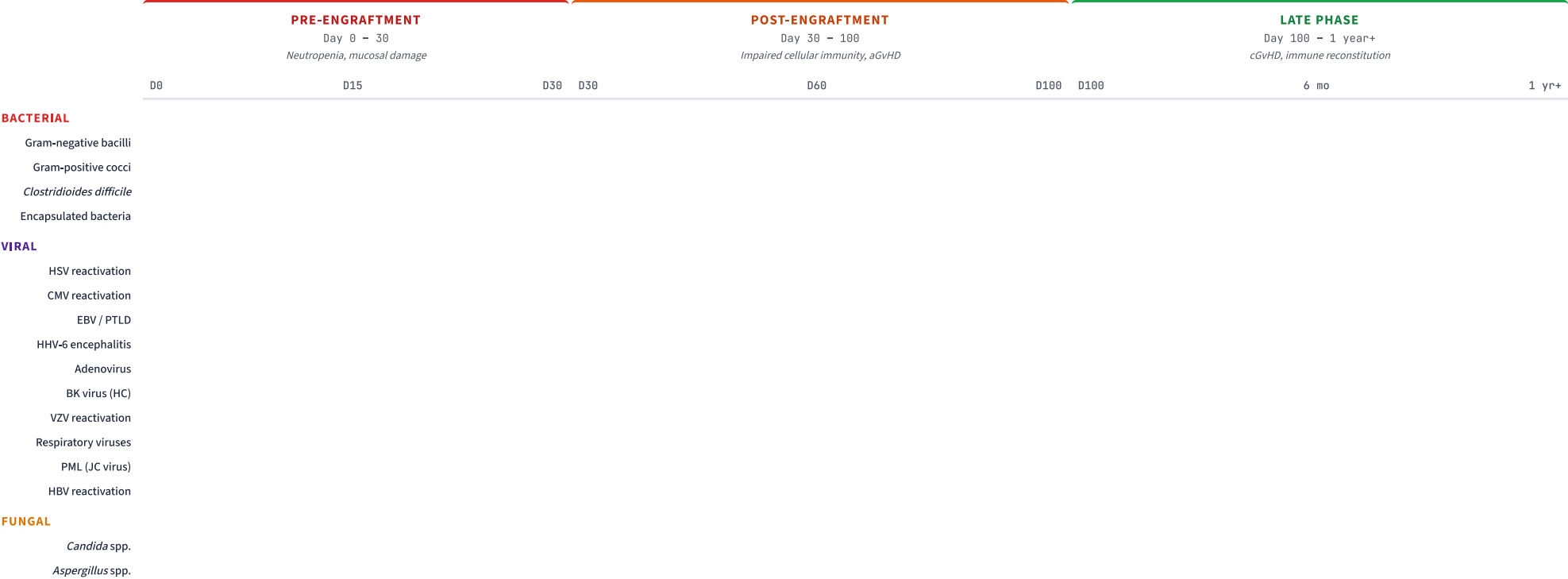
Allogeneic hematopoietic stem cell transplantation (HSCT) recipients



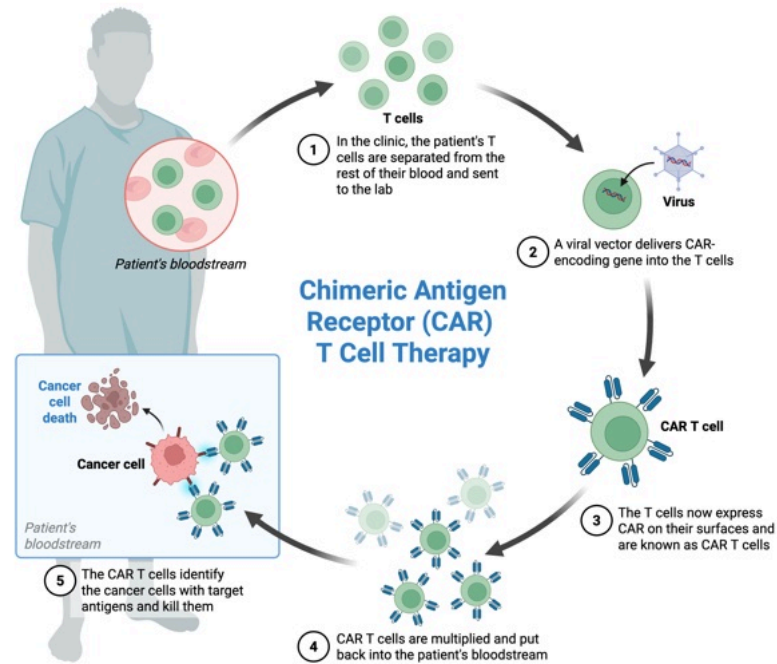
Timeline of Infection Risk

Infections After Allogeneic Hematopoietic Stem Cell Transplantation

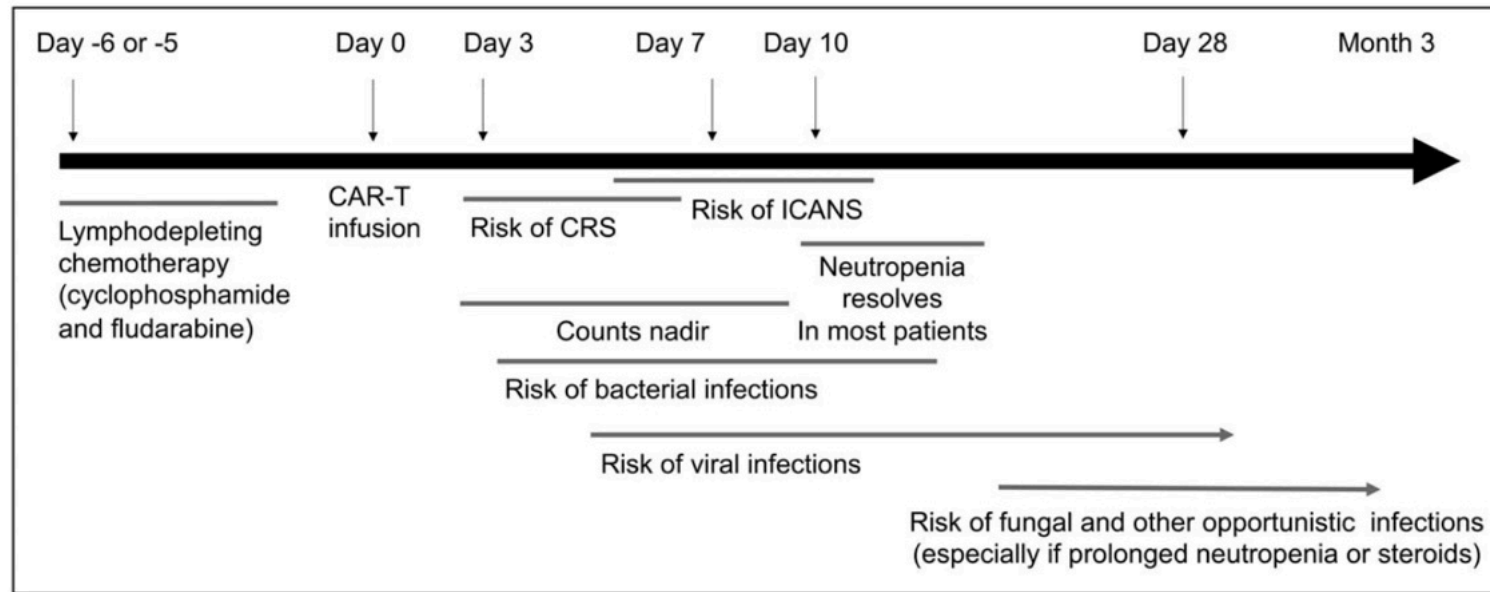
Temporal distribution of infectious complications by pathogen type and transplant phase



What is CAR-T therapy?



Immunosuppression timeline with CAR-T



CAR-T: Chimeric antigen receptor T-cell therapy; CRS-cytokine release syndrome; ICANS-Immune cell-associated neurological syndrome

CAR-T cell therapy

- Patients are **profoundly immunosuppressed**
- Up to 1/3 suffer serious bacterial infection in first 30 days (Stewart and Henden, 2021)
- Cytokine release syndrome complicates assessment
- Prolonged B-cell aplasia → hypogammaglobulinemia

Measuring Immunosuppression

Available Markers

Useful in HIV:

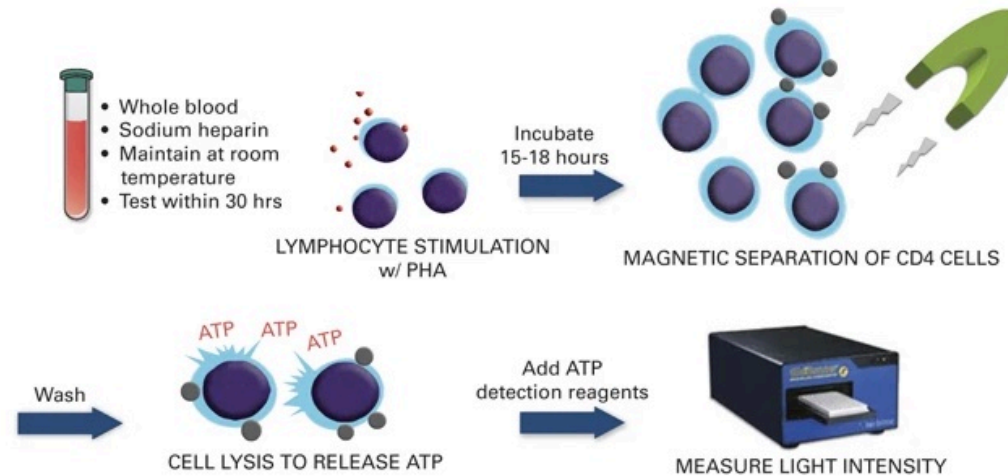
- CD4 count
- CD4 percentage
- CD4/CD8 ratio

General markers:

- Neutrophil count
- Lymphocyte count
- Immunoglobulin levels

Unfortunately, we lack reliable composite markers or algorithms to predict infection risk in most immunocompromised populations outside

Emerging biomarkers



- **Viral reactivation** (EBV, CMV, TTV, BK) → correlates with immunosuppression
- **QuantiFERON Monitor** → may identify over-immunosuppression
- **ImmuKnow assay** → correlates with infection/rejection risk
- Traditional markers (ESR, CRP, procalcitonin) → **NOT predictive**

Sources of Infection

Community-acquired pathogens

- Most common infections **mimic community pathogens**
- Immunocompromised patients are often “sentinel cases” in outbreaks
- Respiratory viruses, GI pathogens
- *Norovirus*, *C. difficile*

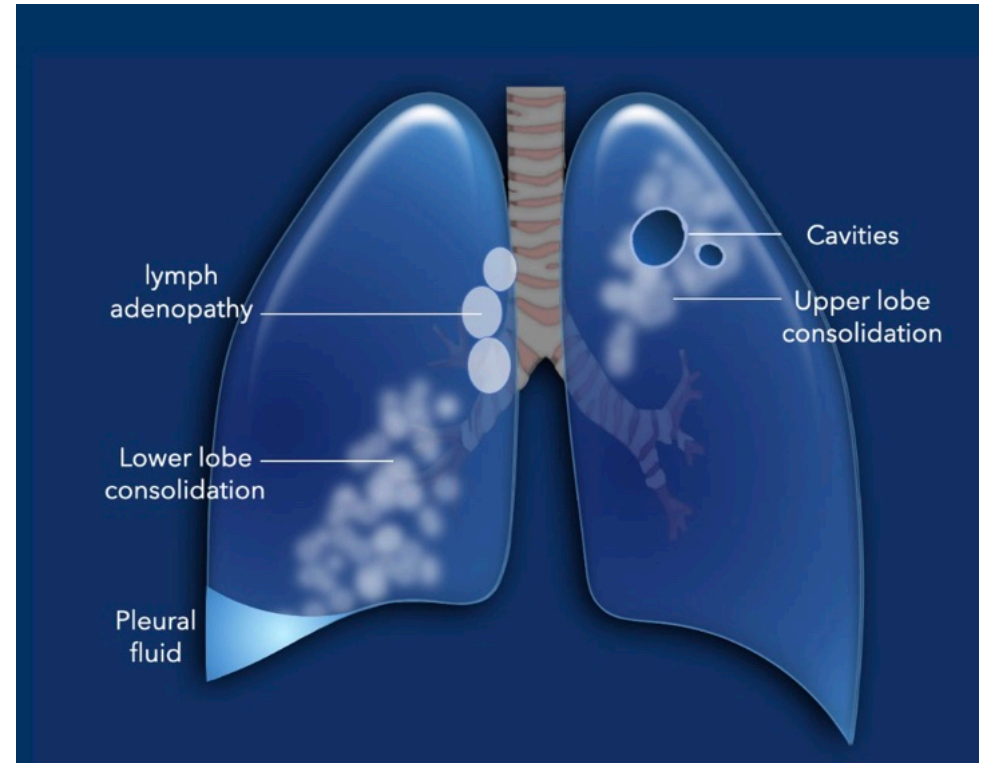
Healthcare-associated pathogens

- **Increased risk of MDR organisms** due to frequent healthcare contact
- Catheter-related infections
- Pneumonia
- UTI

Reactivation of latent infections

Key pathogens to screen for and monitor:

- *Mycobacterium tuberculosis*
- *Strongyloides*
- Hepatitis B
- *Coccidioides*, *Histoplasma*
- *Trypanosoma cruzi* (Chagas)



Donor-derived infections

- Organ transplant
- Stem cell transplant
- Blood products
- Usually within **first 6 months**
- Most common infections:
 - Cytomegalovirus
 - Epstein-Barr virus (post-transplant lymphoproliferative disease)
 - Herpes simplex and varicella zoster
 - Hepatitis B,C
 - HIV
 - Bacterial infections
 - Fungal infections (*Candida*, *Aspergillus*)

Components of Host Defense

Overview of immune system

Innate Immunity

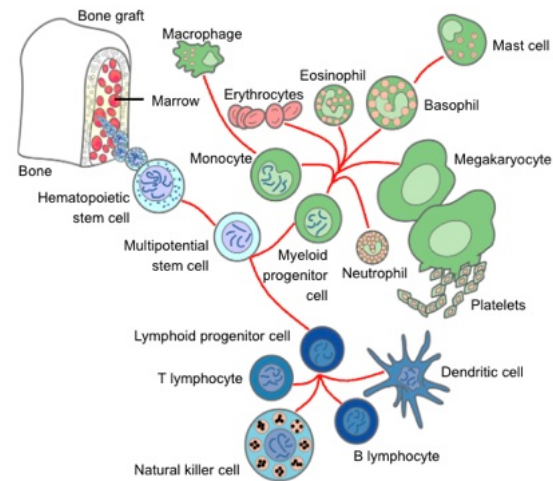
- Granulocytes
- Monocytes/Macrophages
- NK cells
- Complement
- Physical barriers

Acquired Immunity

- Cellular (T cells)
- Humoral (B cells)
- Antibody production

Granulocytes (neutrophils)

- Chemotherapy & radiation → **neutropenia**
- Duration: 3-4 weeks or longer
- **Primary risk factor** for infection
- Risk increases with:
 - Depth of neutropenia
 - Duration of neutropenia
- Concurrent organ dysfunction



Corticosteroid effects on neutrophils

Paradoxical effects:

- ↑ Granulocytopoiesis (apparent benefit)
 - BUT: ↓ Accumulation at infection site
- ↓ Adherent capacity
- ↓ Chemotaxis
- ↓ Phagocytosis
- ↓ Intracellular killing

Monocytes & macrophages

- **Monocytopenia** parallels neutropenia
- Macrophage activation requires **T-cell cytokines** (IFN- γ)
- Explains cellular immunodeficiency susceptibility
- Targeted therapies can have **unexpected effects**

NK cells and platelets

NK Cells:

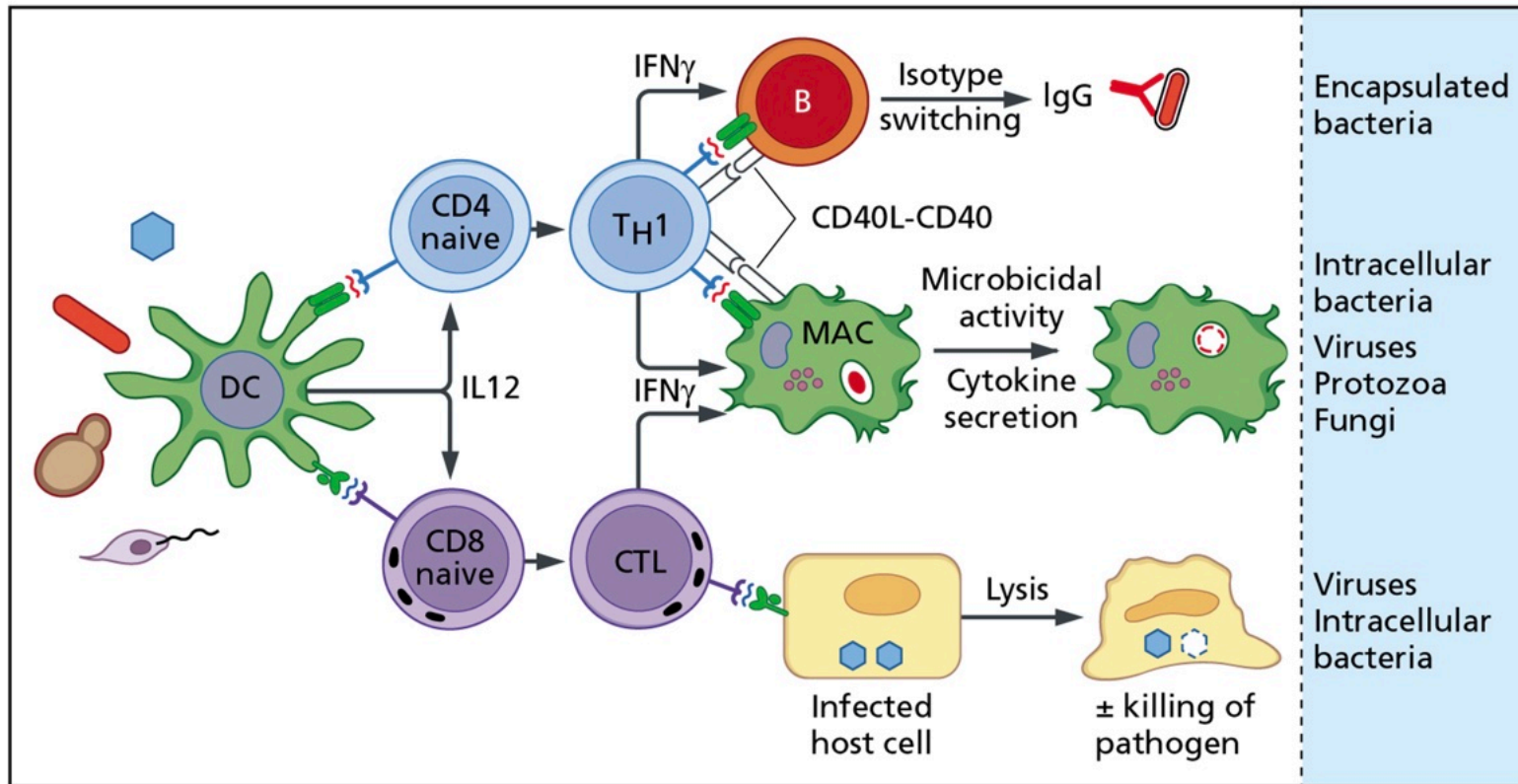
- Respond to viruses and malignancy
- CD56 receptor → *Aspergillus* recognition
- Dysfunction contributes to fungal susceptibility

Platelets:

- Increasingly recognized immune role
- Thrombocytopenia → independent bacteremia risk
- Protection against yeast and molds

Cellular Immunity

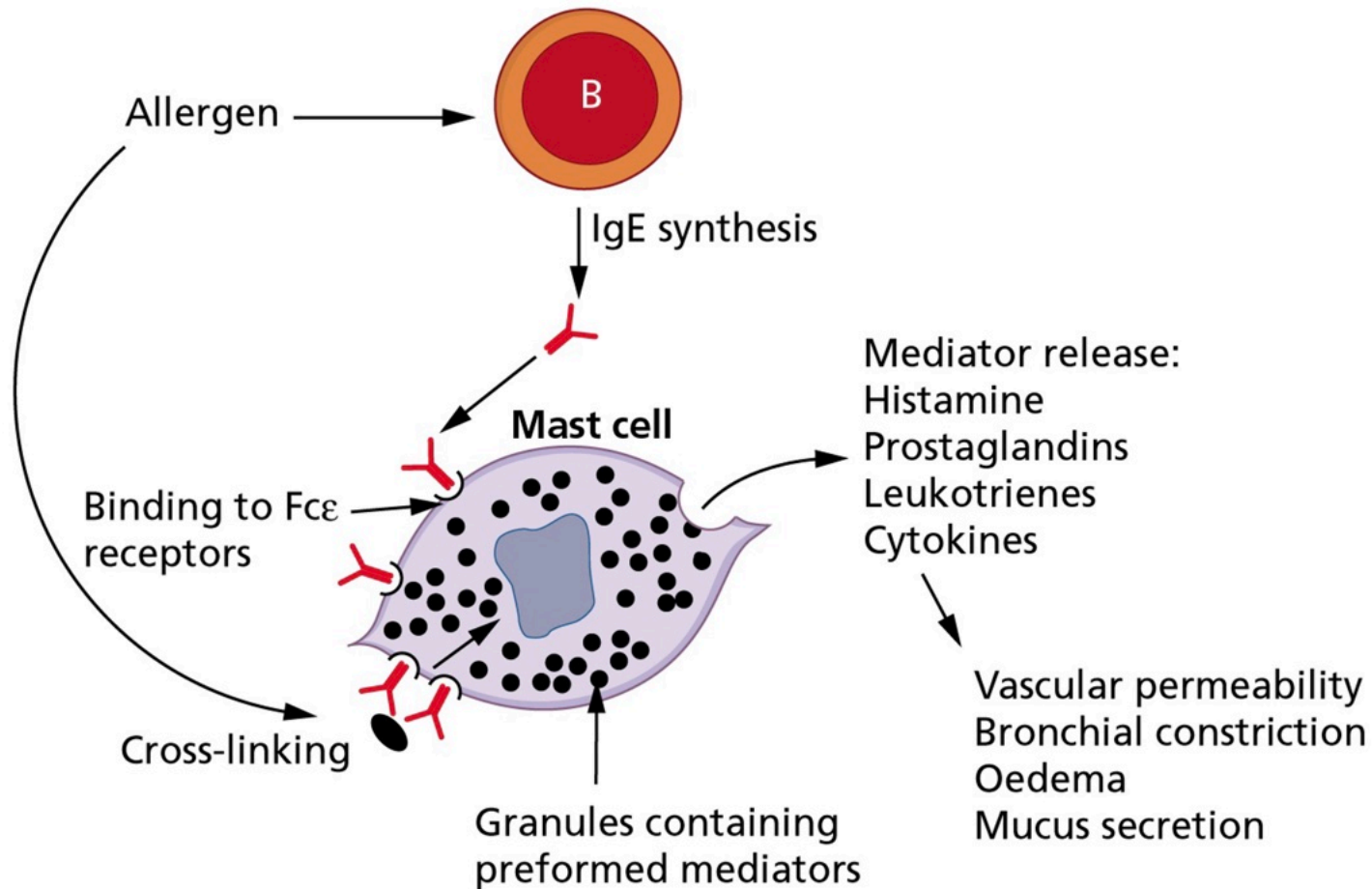
Specific cell-mediated immune defects: Th1 and CTL-driven immunity



Specific cell-mediated immune defects: Th2 and CTL-driven immunity



Cell-mediated driven allergy



Drugs that impair cell-mediated immunity

Common drugs that impair T-cell function:

- Corticosteroids
- Azathioprine, cyclosporine, tacrolimus
- mTOR inhibitors (sirolimus, everolimus)
- Purine analogues (fludarabine, cladribine)
- Alemtuzumab

Diseases: Hodgkin lymphoma, CLL

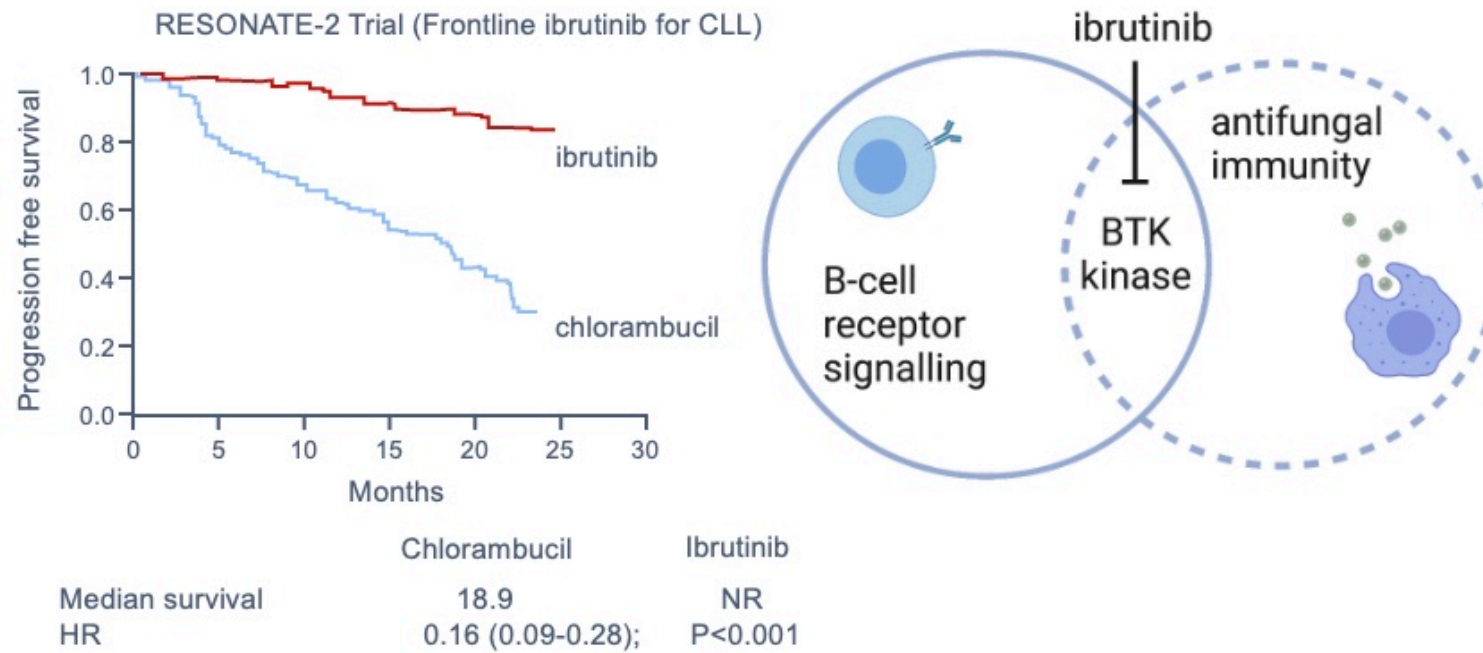
Examples of “Targeted therapy” risks

Drug	Mechanism	Infection Risk
Ruxolitinib	JAK-STAT inhibitor	TB, HBV reactivation
Ibrutinib	BTK inhibitor	Aspergillosis, PJP
Idelalisib	PI3K inhibitor	<i>P. jirovecii</i>

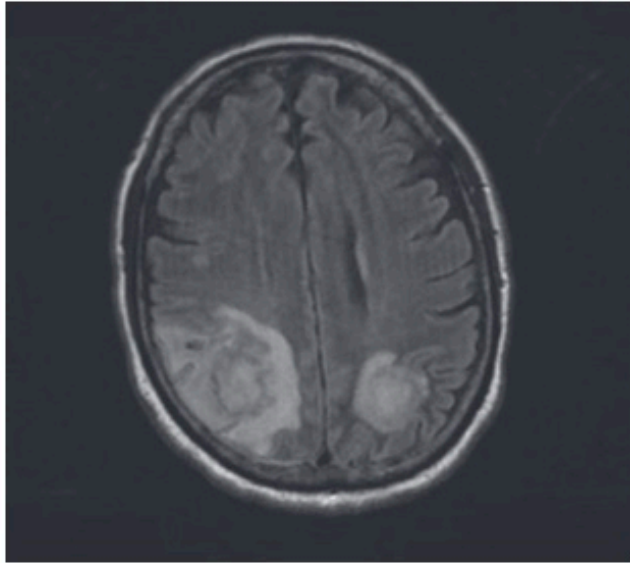
If you see a drug ending in “mab” or “nib” or “sib”consider unique infection risk

mab- monoclonal antibody-large protein biologic usually IV that binds extracellular targets (receptors, cytokines, cells); nib-small molecule

Ibrutinib



Unexpected fungal infections



T2-weighted FLAIR magnetic resonance imaging revealing multiple cerebral abscesses.

- Retrospective French surveillance (FILO) identified 33 cases of aspergillosis in ibrutinib-treated patients with chronic lymphocytic leukemia
- **CNS aspergillosis in 11/27 (40%),** most cases within 3 months of starting ibrutinib therapy
- All patients had refractory/ relapsed disease and additional predisposing factors in most of patients (e.g., steroids in 7, neutropenia in 5)

Infectious Complications & Mortality

Humoral immunity

- **B cells** → antibody-secreting plasma cells
- Impaired in CLL, multiple myeloma
- Rituximab, CAR-T → **B-cell depletion**
- Profound, long-lasting hypogammaglobulinemia

Splenectomy: Loss of encapsulated bacteria defense-“Big 3”

- *Streptococcus pneumoniae*
- *Haemophilus influenzae* type B
- *Neisseria meningitidis*

Less common: *Capnocytophaga canimorsus*, *Salmonella* spp. *E. coli*

PSV and PPSV23 vaccine, MENACWY and MenB vaccine, HIB, Influenzae- Patients should receive vaccines 2 weeks before elective splenectomy or 2 weeks after emergency splenectomy

Physical Barriers

The Integument

Skin:

- Chemotherapy → hair loss, dryness
- Catheters → direct microbial access
- Broken skin → *S. aureus*, gram-negatives



Oropharynx:

- Xerostomia + antibiotics → thrush, bacterial overgrowth

Alimentary Tract

Grade 1:
Soreness and
Erythema



Grade 2:
Erythema and ulcers,
Patient can swallow food



Grade 3
Ulcers with extensive
Erythema;
Cannot swallow food



Grade 4
Alimentation is not
Possible



- **Microbiome disruption** with antibiotics → *C. difficile*
- Mucosal barrier injury from chemotherapy
- Facilitates bacterial translocation
- With concomitant neutropenia allows progression to sepsis

Immunodeficiency-Pathogen Associations

Neutropenia: Gram-positive pathogens

- Coagulase-negative staphylococci more common than *Staphylococcus aureus* (most are from central venous catheter)
- Viridans streptococci
- Enterococci

Neutropenia: Gram-negative pathogens

- *Escherichia coli*
- *Pseudomonas aeruginosa*
- *Klebsiella pneumoniae*
- *Enterobacter* spp.

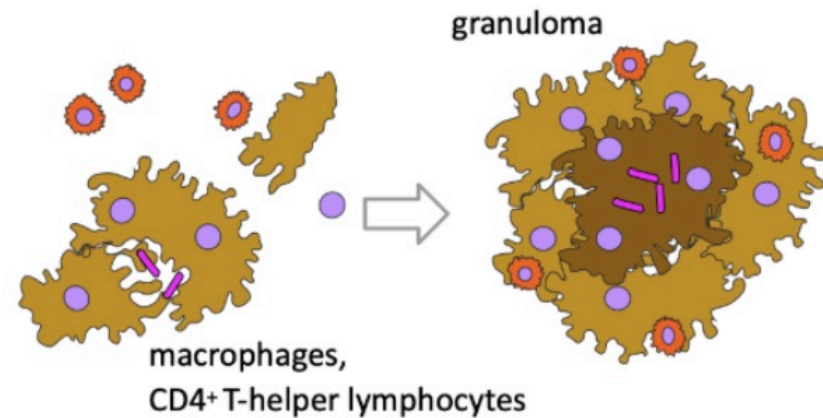
Impaired cellular immunity

Bacteria/Mycobacteria:

- *Listeria monocytogenes*
- *Nocardia* spp.
- *M. tuberculosis*
- Nontuberculous mycobacteria

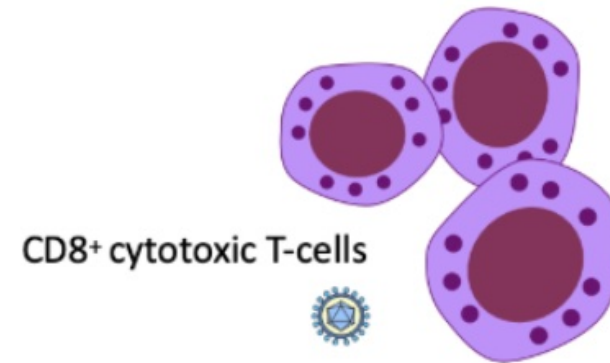
Fungi/Parasites:

- *P. jirovecii*
- *Aspergillus* spp.
- *Cryptococcus* spp.
- *Toxoplasma gondii*



Impaired cellular immunity (viruses)

- Herpesviruses (HSV, VZV, CMV, EBV)
- Respiratory viruses
- Polyomaviruses (BK, JC)
- Human papillomavirus



Impaired humoral immunity

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Neisseria meningitidis*
- Norovirus
- Hepatitis B virus



Prevention Strategies

Pre-immunosuppression screening

Screen before starting immunosuppressive therapy:

Pathogen	Test	Action if Positive
<i>M. tuberculosis</i>	IGRA (QuantiFERON) or TST	Isoniazid or rifampin prophylaxis
Hepatitis B	HBsAg, anti-HBc, anti-HBs	Antiviral prophylaxis (entecavir/tenofovir)
Hepatitis C	Anti-HCV, HCV RNA	Treat prior to immunosuppression if possible
HIV	4th-gen Ag/Ab	ART optimization
<i>Strongyloides</i>	Serology (endemic areas/travel)	Ivermectin × 2 doses
<i>T. cruzi</i>	Serology (Latin American origin)	Benznidazole prophylaxis
VZV	IgG serology	Vaccinate if seronegative (pre-therapy)
<i>Coccidioides/Histoplasma</i>	Serology (endemic exposure)	Antifungal prophylaxis

Prophylaxis principles

TMP-SMX for PJP :typically 1 DS tablet daily also covers:

- *Toxoplasma*
- *S. aureus*
- *Nocardia*
- Many gram-positives/negatives

Antiviral prophylaxis: Val(acyclovir) for CMV (weak activity), HSV, VZV prevention. Valganciclovir or letermovir for higher risk CMV patients

Problem with TMP/SMX- rash, GI upset, and cytopenias—often limits adherence. Can use - 1 DS tablet 3×/week (e.g., Mon–Wed–Fri) or even single-strength daily. Add folic acid if patient has anemia or leukopenia. Valganciclovir is associated with dose-limiting myelosuppression.

Antifungal prophylaxis

Risk-stratified approach:

High-risk indications (mold-active prophylaxis):

- AML induction/consolidation chemotherapy
- Allogeneic HSCT recipients
- Active GVHD on high-dose steroids
- Lung transplant recipients

Agents of choice: - Posaconazole (oral suspension or tablet) - Voriconazole (alternative) - Micafungin (IV, if oral not tolerated)

Lower-risk indications (Candida prophylaxis only):

- Autologous HSCT
- Prolonged neutropenia (>7 days)
- Mucositis + broad-spectrum antibiotics

Agents of choice: - Fluconazole - Micafungin

HBV reactivation prevention

A critical and underrecognized risk:

- HBV reactivation can occur with **any immunosuppressive regimen**
- Risk highest with: rituximab, anti-CD20 agents, anthracycline chemotherapy, corticosteroids ≥ 20 mg/day for ≥ 4 weeks, stem cell transplant

Who to treat:

- HBsAg(+): **always** give prophylaxis
- HBsAg(-)/anti-HBc(+): prophylaxis if high-risk regimen
- Reactivation mortality: **up to 25%** without prophylaxis

Prophylaxis options:

- **Entecavir** (preferred — high barrier to resistance)
- **Tenofovir** (TDF or TAF)
- Duration: continue **6–12 months after** cessation of immunosuppression (18–24 months for anti-CD20)

Preemptive vs. prophylactic strategies for CMV

Two complementary approaches:

Prophylaxis (preferred when risk is high):

- Universal antiviral drug for defined at-risk period
- **Letermovir** (CMV D+/R- or R+ allo-HSCT)
- **Valganciclovir** (SOT, especially lung/heart recipients)
- Advantage: prevents CMV disease directly
- Disadvantage: late-onset CMV, drug toxicity, cost

Preemptive therapy (intermediate risk):

- Regular **CMV PCR surveillance** (every 1–2 weeks)
- Treat when viral load crosses threshold before disease
- Advantage: less antiviral exposure
- Disadvantage: requires rigorous monitoring; may miss rapid progressors

Choice of strategy depends on donor/recipient serostatus, transplant type, and institutional resources. D+/R- allo-HSCT patients have

Vaccination: Solid Organ Transplant recipients

Core principle: Vaccinate **before** transplant whenever possible — responses are significantly blunted post-transplant on maintenance immunosuppression.

Vaccine	Pre-transplant	Post-transplant	Notes
Influenza (inactivated)	✅ Annually	✅ Annually (≥1 month post-Tx)	Live attenuated influenza: contraindicated
Pneumococcal (PCV20)	✅ ≥2 weeks before	✅ ≥3–6 months post-Tx	Booster PPSV23 at 5 years if PCV15 series used
COVID-19 (mRNA)	✅	✅ ≥1 month post-Tx; extra doses often needed	Check serology — responses frequently inadequate
Hepatitis B (double-dose)	✅ Check anti-HBs	✅ If not immune; confirm with anti-HBs	Anti-HBs target ≥10 IU/L
Tdap / Td	✅	✅ ≥6 months post-Tx	Pertussis booster every 10 years
MMR, VZV (live)	✅ ≥4 weeks before	❌ Contraindicated post-transplant	Vaccinate seronegative candidates pre-listing
HPV	✅ (if age-eligible)	✅ ≥6 months post-Tx	3-dose series; may respond poorly

Household contacts and close caregivers should receive annual influenza (inactivated), MMR, and varicella vaccines to create a “cocoon” of protection. Verify VZV and measles serostatus in all transplant

Vaccination: Hematopoietic stem cell transplant recipients

Core principle: HSCT ablates immune memory — recipients must be **fully revaccinated** post-transplant regardless of pre-transplant history.

Vaccine	Timing Post-HSCT	Notes
Influenza (inactivated)	≥6 months (or ≥4 months during seasonal outbreak)	Annual; live influenza contraindicated
Pneumococcal (PCV20/PCV15 × 3, then PPSV23)	Start at 3–6 months	3-dose PCV series then PPSV23 ≥8 weeks later
COVID-19 (mRNA)	≥3–6 months; 3-dose primary series	Check serology; additional doses often needed in GVHD
Hepatitis B (double-dose × 3)	≥6 months	Confirm anti-HBs ≥10 IU/L; re-dose if inadequate
Tdap, then Td boosters	≥6 months	3-dose diphtheria/tetanus/pertussis series
Inactivated polio (IPV)	≥6 months	3-dose series; OPV contraindicated
Haemophilus influenzae b (Hib)	≥6 months	3-dose series; covers encapsulated bacteria
Meningococcal (ACWY + B)	≥6 months	Important in asplenic/hyposplenic post-HSCT
MMR, VZV (live)	≥24 months post-HSCT	Only if: off all immunosuppression, no active GVHD, CD4 ≥200

Revaccination should be **delayed** during active GVHD or while on immunosuppressive therapy beyond standard prophylaxis. Serologic testing 4–8 weeks after vaccination helps confirm adequate

Immunoglobulin replacement therapy

Indications:

- Severe hypogammaglobulinemia (IgG <400 mg/dL) with recurrent infections
- Post-rituximab or CAR-T B-cell aplasia
- Common variable immunodeficiency (CVID)
- Post-HSCT (selected patients)

Products:

- **IVIG** (intravenous): 0.4–0.6 g/kg every 3–4 weeks
- **SCIG** (subcutaneous): home infusion, more stable IgG trough levels

Monitoring:

- Target trough IgG >500–700 mg/dL
- Adjust dose based on infection breakthrough
- Duration: until immune reconstitution (may be 1–3 years)

SCIG is increasingly preferred for patient convenience and more stable pharmacokinetics. Cost and insurance coverage remain barriers in

Environmental & infection control measures

Hospital setting (high-risk patients):

- **HEPA-filtered rooms** with positive pressure for prolonged neutropenia and allo-HSCT
- Avoid construction/renovation zones (aerosolized *Aspergillus*)
- Strict hand hygiene — single most effective infection prevention measure
- Central line bundles to reduce CLABSI

Dietary precautions (neutropenic diet — evidence limited but commonly practiced):

- Avoid raw/undercooked meat, eggs, unpasteurized products
- Wash fruits and vegetables thoroughly
- No well-ripened soft cheeses (*Listeria*)

The evidence base for strict “neutropenic diet” is weak; current IDSA/ASBMT guidelines suggest emphasis on food safety practices rather

Patient education

High-risk exposures to avoid:

- Gardening without protection (molds, *Nocardia*)
- Poor dental hygiene (*Actinomyces*, bacteremia)
- Marijuana smoking (*Aspergillus*)
- Raw seafood (*Vibrio*)
- Warm ocean swimming

Prophylaxis by immunodeficiency type

Defect	Representative Conditions	Antibacterial	Antifungal	Antiviral	Other
Neutropenia (profound, >7d)	AML induction, allo-HSCT engraftment	Levofloxacin†	Posaconazole (high-risk) / fluconazole	Acyclovir (HSV)	G-CSF if prolonged
T-cell deficiency	Allo-HSCT, calcineurin inhibitors, alemtuzumab, CAR-T	TMP-SMX (PJP, <i>Nocardia</i> , <i>Toxoplasma</i>)	Posaconazole / voriconazole	Letermovir or valganciclovir (CMV); acyclovir (HSV/VZV)	Screen for LTBI, <i>Strongyloides</i> ; CMV surveillance
Humoral / B-cell deficiency	Rituximab, CAR-T, CLL, myeloma	TMP-SMX (PJP); consider azithromycin	Not routinely required	Acyclovir; monitor for enterovirus	IVIG if IgG <400 mg/dL + recurrent infections
Hypogammaglobulinemia	CVID, post-HSCT (late phase)	TMP-SMX or azithromycin	Not routinely required	Acyclovir	IVIG/SCIG replacement; PCV20, HIB, MenACWY vaccines
Asplenia / functional hyposplenism	Splenectomy, sickle cell, splenic irradiation	Penicillin V / amoxicillin (lifelong in many)	Not routinely required	Not routinely required	PCV20 + PPSV23, MenACWY + MenB, HIB, annual influenza; emergency antibiotic supply

Key Takeaways

Summary points

1. **6% of population** is immunocompromised
2. **Net state of immunosuppression** = composite assessment
3. **First 100 days after transplant** = highest risk period
4. **No single marker** predicts infection risk
5. **Know pathogen associations** with specific defects
6. **Prophylaxis** significantly alters risk profile

Clinical Pearls

Remember

- TMP-SMX provides broader coverage than just PJP prophylaxis
- Timing matters—early vs late infections differ
- Targeted therapies (-mabs -nibs) have unexpected infection risks
- Consider the whole patient, not just the lab values

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